Lecture 32: Normal Forms — 1NF, 2NF, 3NF, BCNF

Z2004: Database Management Systems

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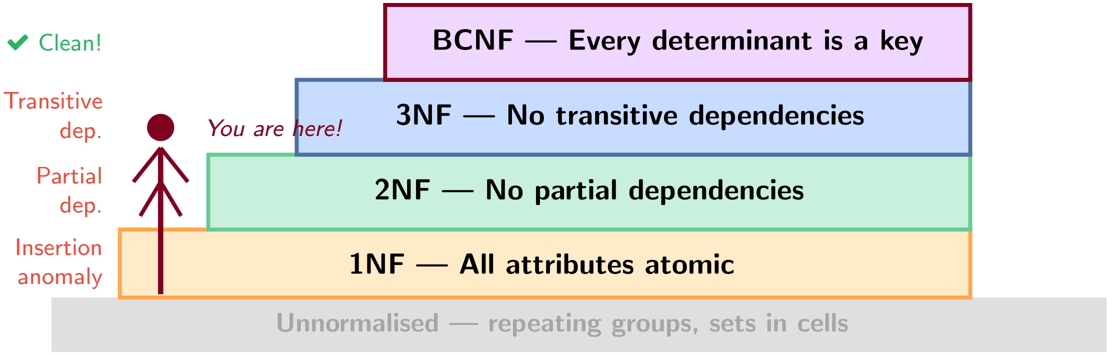
Outline

- The normalisation staircase
- 1NF: Atomic attributes
- 2NF: No partial dependencies
- 3NF: No transitive dependencies
- BCNF: Every determinant is a key
- 3NF vs. BCNF trade-off
- BCNF decomposition algorithm
- End-to-end worked example
- Think-Pair-Share + Summary

Anchor Visual

We return to our messy `StudentEnrolment` table from L30 and normalise it step by step to BCNF.

The Normalisation Staircase



1NF — First Normal Form: Atomic Attributes

✖ BAD: Non-atomic column

SID	SName	PhoneNumbers
S001	Amani	**0712111222, 0723333444**
S002	Baraka	**0711555666**
S003	Chidi	**0700888999, 0712000111**

Problems:

- How to search for one number?
- How to add a third number?
- Joins are impossible

✔ GOOD: Separate Phone table

SID	Phone
S001	0712111222
S001	0723333444
S002	0711555666
S003	0700888999
S003	0712000111

2NF — No Partial Dependencies

✖ BAD: Partial dependency

Composite key: {**StudentID, CourseID**}

SID	CID	CName	Grade
S001	CS101	**Databases**	A
S002	CS101	**Databases**	B
S003	CS202	**Algorithms**	A

CourseName depends only on CourseID — partial dependency!

✔ GOOD: Decomposed

CID	CName
CS101	Databases
CS202	Algorithms

SID	CID	Grade
S001	CS101	A
S002	CS101	B
S003	CS202	A

3NF — No Transitive Dependencies

✖ BAD: Transitive dependency

SID	SName	DeptID	DeptName
S001	Amani	D01	**CS Dept**
S002	Baraka	D01	**CS Dept**
S003	Chidi	D02	**EE Dept**

$$\begin{tikzcd} \text{SID} \arrow[r] \arrow[rr, bend right, dashed, "\textbf{transitive!}"'] & \text{DeptID} \arrow[r] & \text{DeptName} \end{tikzcd}$$

✔ GOOD: Separate Dept table

DeptID	DeptName
D01	CS Dept
D02	EE Dept

SID	SName	DeptID
S001	Amani	D01
S002	Baraka	D01
S003	Chidi	D02

DeptName stored once.

BCNF — Boyce-Codd Normal Form

BCNF Rule

For every non-trivial FD $X \to Y$: X **must be a candidate key.**

Stricter than 3NF. In 3NF, Y could be part of a candidate key. BCNF: no exceptions.

Classic BCNF violation example:

`Teaching(StudentID, Subject, Instructor)`

- Each student takes each subject from **one** instructor
- Each instructor teaches only **one** subject
- Candidate keys: {SID, Subject} and {SID, Instructor}

✖ 3NF but

"Dr. K
stored

Instruct
Stud

3NF vs. BCNF — The Trade-off

Property	3NF	BCNF	Notes
Lossless-join decomposition	✔ Always	✔ Always	Both guarantee no spurious rows
Dependency preservation	✔ Always	⚠ Sometimes not	BCNF may lose some FDs
Anomaly elimination	Partial	✔ Complete	BCNF removes all redundancy
Achievability	✔ Always possible	✔ Always possible	Both decompositions exist

Use BCNF when…

- Anomaly-free storage is priority
- Dependency checking via joins is acceptable
- Typical OLTP transactional systems

Use 3NF when…

- Dependency preservation is essential
- Cannot afford joins to check FDs
- Multiple overlapping candidate keys exist

Rule of Thumb

Try BCNF first. If a dependency is lost, fall back to 3NF.

BCNF Decomposition Algorithm

Algorithm

1. **Input:** schema R, FD set F
2. If R is in BCNF, **done**
3. Find a non-BCNF FD $\alpha \rightarrow \beta$ (i.e. α is not a key)
4. Decompose R into:
   - $R_1 = \alpha \cup \beta$
   - $R_2 = R - \beta$ (keep α)
5. Recursively check R_1 and R_2

Worked Example:

$R(\underline{A}, B, C, D)$, FDs: $A \rightarrow BCD$, $B \rightarrow C$

$R(A, B, C, D)$

$B \rightarrow C$ violates BCNF

decompose on $B \rightarrow C$

$R_1(B, C)$

✔ BCNF

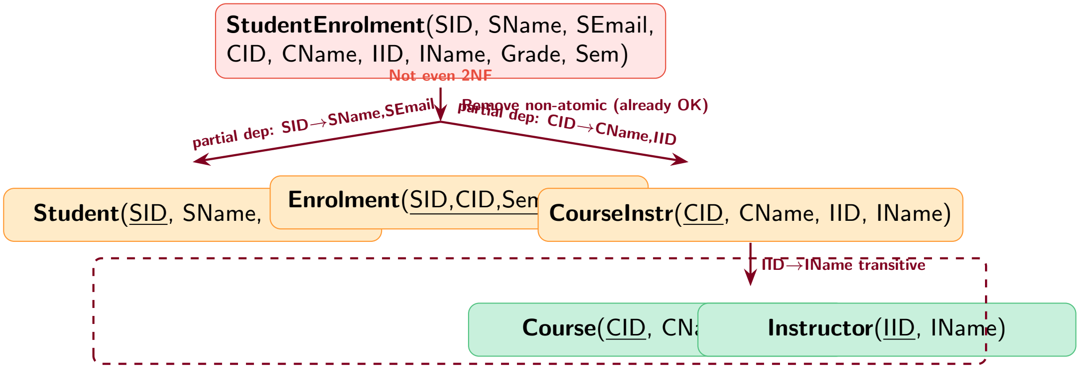
$R_2(A, B, D)$

✔ BCNF (A is key)

Key Properties

BCNF decomposition is always **lossless-join**. It may or may not preserve all FDs. Check after each step.

End-to-End: Normalising `StudentEnrolment` to BCNF



✔ **All schemas now in BCNF**

Think-Pair-Share (5 minutes)

Normalise this schema to BCNF

Schema:

`HospitalVisit(PatientID, PatientName, DoctorID, DoctorName, Specialty, RoomNo, VisitDate, Diagnosis)`

FDs:

- PatientID $\rightarrow$ PatientName
- DoctorID $\rightarrow$ DoctorName, Specialty
- RoomNo $\rightarrow$ DoctorID
- $\{$PatientID, VisitDate$\}$ $\rightarrow$ DoctorID, RoomNo, Diagnosis

- What are the candidate keys?
- Which FD violates BCNF first?
- How many schemas in your final result?
- Any lost FDs?

Think

RoomNo $\rightarrow$ DoctorID: is RoomNo a candidate key?

Summary

Normal Form	Requires	Eliminates	Key FD Concept
1NF	Atomic values	Repeating groups	—
2NF	1NF + no partial deps	Partial dependency	Part of key → non-key
3NF	2NF + no transitive deps	Transitive dependency	Non-key → non-key
BCNF	Every determinant is a key	All FD-based anomalies	$X \to Y$ with X not a key

The Big Picture

- Higher NF = fewer anomalies
- Each step targets a specific problem
- BCNF is the "gold standard" for FDs
- MVDs need 4NF (next lecture)

Next Lecture — L33: 4NF & 5NF

Beyond FDs: **Multi-valued dependencies** (MVDs). A BCNF table can *still* have redundancy. 4NF fixes this.